vignesh07 / clawdbot-railway-template

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Public repository

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vignesh07 chore: bump OPENCLAW_GIT_REF to v2026.3.8 (#177) 49a5613 · last week

.github/workflows	chore: add action to bump OPENCL...	last month
assets	docs(readme): add Railway official +...	last month
scripts	chore: rename package/links back t...	last month
src	fix: exempt /hooks/* from requireD...	2 weeks ago
test	fix(ws): don't require Basic auth on ...	3 weeks ago
.gitignore	Initial Railway template with /setup ...	2 months ago
CONTRIBUTING.md	chore: rename package/links back t...	last month
Dockerfile	chore: bump OPENCLAW_GIT_REF t...	last week
LICENSE	Initial commit	2 months ago
README.md	fix: add dashboard auth + token syn...	3 weeks ago
package-lock.json	chore: rename package/links back t...	last month
package.json	chore: rename package/links back t...	last month
railway.toml	Railway template: respect injected P...	last month

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OpenClaw Railway Template (1-click deploy)

This repo packages **OpenClaw** for Railway with a small **/setup** web wizard so users can deploy and onboard **without running any commands**.

What you get

- **OpenClaw Gateway + Control UI** (served at `/` and `/openclaw`)
- A friendly **Setup Wizard** at `/setup` (protected by a password)
- Persistent state via **Railway Volume** (so config/credentials/memory survive redeploys)
- One-click **Export backup** (so users can migrate off Railway later)
- **Import backup** from `/setup` (advanced recovery)

How it works (high level)

- The container runs a wrapper web server.
- The wrapper protects `/setup` (and the Control UI at `/openclaw`) with `SETUP_PASSWORD` using HTTP Basic auth.
- During setup, the wrapper runs `openclaw onboard --non-interactive ...` inside the container, writes state to the volume, and then starts the gateway.
- After setup, **`/` is OpenClaw**. The wrapper reverse-proxies all traffic (including WebSockets) to the local gateway process.

Railway deploy instructions (what you'll publish as a Template)

In Railway Template Composer:

1. Create a new template from this GitHub repo.
2. Add a **Volume** mounted at `/data`.
3. Set the following variables:

Required:

- `SETUP_PASSWORD` — user-provided password to access `/setup` and the Control UI (`/openclaw`) via HTTP Basic auth

Recommended:

- `OPENCLAW_STATE_DIR=/data/.openclaw`
- `OPENCLAW_WORKSPACE_DIR=/data/workspace`

Optional:

- `OPENCLAW_GATEWAY_TOKEN` — if not set, the wrapper generates one (not ideal). In a template, set it using a generated secret.

Notes:

- This template pins OpenClaw to a released version by default via Docker build arg `OPENCLAW_GIT_REF` (override if you want `main`).

4. Enable **Public Networking** (HTTP). Railway will assign a domain.
   - This service listens on Railway's injected `PORT` at runtime (recommended).
5. Deploy.

Then:

- Visit `https://<your-app>.up.railway.app/setup`
  - Your browser will prompt for **HTTP Basic auth**. Use any username; the password is `SETUP_PASSWORD`.
- Complete setup
- Visit `https://<your-app>.up.railway.app/` and `/openclaw` (same Basic auth)

Support / community

- GitHub Issues: https://github.com/vignesh07/clawdbot-railway-template/issues
- Discord: https://discord.com/invite/clawd

If you're filing a bug, please include the output of:

- `/healthz`
- `/setup/api/debug` (after authenticating to /setup)

Getting chat tokens (so you don't have to scramble)

Telegram bot token

1. Open Telegram and message **@BotFather**
2. Run `/newbot` and follow the prompts
3. BotFather will give you a token that looks like: `123456789:AA...`
4. Paste that token into `/setup`

Discord bot token

1. Go to the Discord Developer Portal: https://discord.com/developers/applications
2. **New Application** → pick a name
3. Open the **Bot** tab → **Add Bot**
4. Copy the **Bot Token** and paste it into `/setup`
5. Invite the bot to your server (OAuth2 URL Generator → scopes: `bot`, `applications.commands`; then choose permissions)

Persistence (Railway volume)

Railway containers have an ephemeral filesystem. Only the mounted volume at `/data` persists across restarts/redeploys.

What persists cleanly today:

- **Custom skills / code:** anything under `OPENCLAW_WORKSPACE_DIR` (default: `/data/workspace`)

- **Node global tools (npm/pnpm):** this template configures defaults so global installs land under `/data`:
  - npm globals: `/data/npm` (binaries in `/data/npm/bin`)
  - pnpm globals: `/data/pnpm` (binaries) + `/data/pnpm-store` (store)
- **Python packages:** create a venv under `/data` (example below). The runtime image includes Python + venv support.

What does *not* persist cleanly:

- `apt-get install ...` (installs into `/usr/*`)
- Homebrew installs (typically `/opt/homebrew` or similar)

Optional bootstrap hook

If `/data/workspace/bootstrap.sh` exists, the wrapper will run it on startup (best-effort) before starting the gateway. Use this to initialize persistent install prefixes or create a venv.

Example `bootstrap.sh`:

```
#!/usr/bin/env bash
set -euo pipefail

# Example: create a persistent python venv
python3 -m venv /data/venv || true

# Example: ensure npm/pnpm dirs exist
mkdir -p /data/npm /data/npm-cache /data/pnpm /data/pnpm-store
```

Troubleshooting

“disconnected (1008): pairing required” / dashboard health offline

This is not a crash — it means the gateway is running, but no device has been approved yet.

Fix:

- Open `/setup`
- Use the **Debug Console**:
  - `openclaw devices list`
  - `openclaw devices approve <requestId>`

If `openclaw devices list` shows no pending request IDs:

- Make sure you’re visiting the Control UI at `/openclaw` (or your native app) and letting it attempt to connect
  - Note: the Railway wrapper now proxies the gateway and injects the auth token automatically, so you should not need to paste the gateway token into the Control UI when using `/openclaw`.
- Ensure your state dir is the Railway volume (recommended): `OPENCLAW_STATE_DIR=/data/.openclaw`
- Check `/setup/api/debug` for the active state/workspace dirs + gateway readiness

“unauthorized: gateway token mismatch”

The Control UI connects using `gateway.remote.token` and the gateway validates `gateway.auth.token`.

Fix:

- Re-run `/setup` so the wrapper writes both tokens.
- Or set both values to the same token in config.

“Application failed to respond” / 502 Bad Gateway

Most often this means the wrapper is up, but the gateway can’t start or can’t bind.

Checklist:

- Ensure you mounted a **Volume** at `/data` and set:
  - `OPENCLAW_STATE_DIR=/data/.openclaw`
  - `OPENCLAW_WORKSPACE_DIR=/data/workspace`
- Ensure **Public Networking** is enabled (Railway will inject `PORT`).
- Check Railway logs for the wrapper error: it will show `Gateway not ready:` with the reason.

Legacy CLAWDBOT_* env vars / multiple state directories

If you see warnings about deprecated `CLAWDBOT_*` variables or state dir split-brain (e.g. `~/.openclaw` vs `/data/...`):

- Use `OPENCLAW_*` variables only
- Ensure `OPENCLAW_STATE_DIR=/data/.openclaw` and `OPENCLAW_WORKSPACE_DIR=/data/workspace`
- Redeploy after fixing Railway Variables

Build OOM (out of memory) on Railway

Building OpenClaw from source can exceed small memory tiers.

Recommendations:

- Use a plan with **2GB+ memory**.
- If you see `Reached heap limit Allocation failed - JavaScript heap out of memory`, upgrade memory and redeploy.

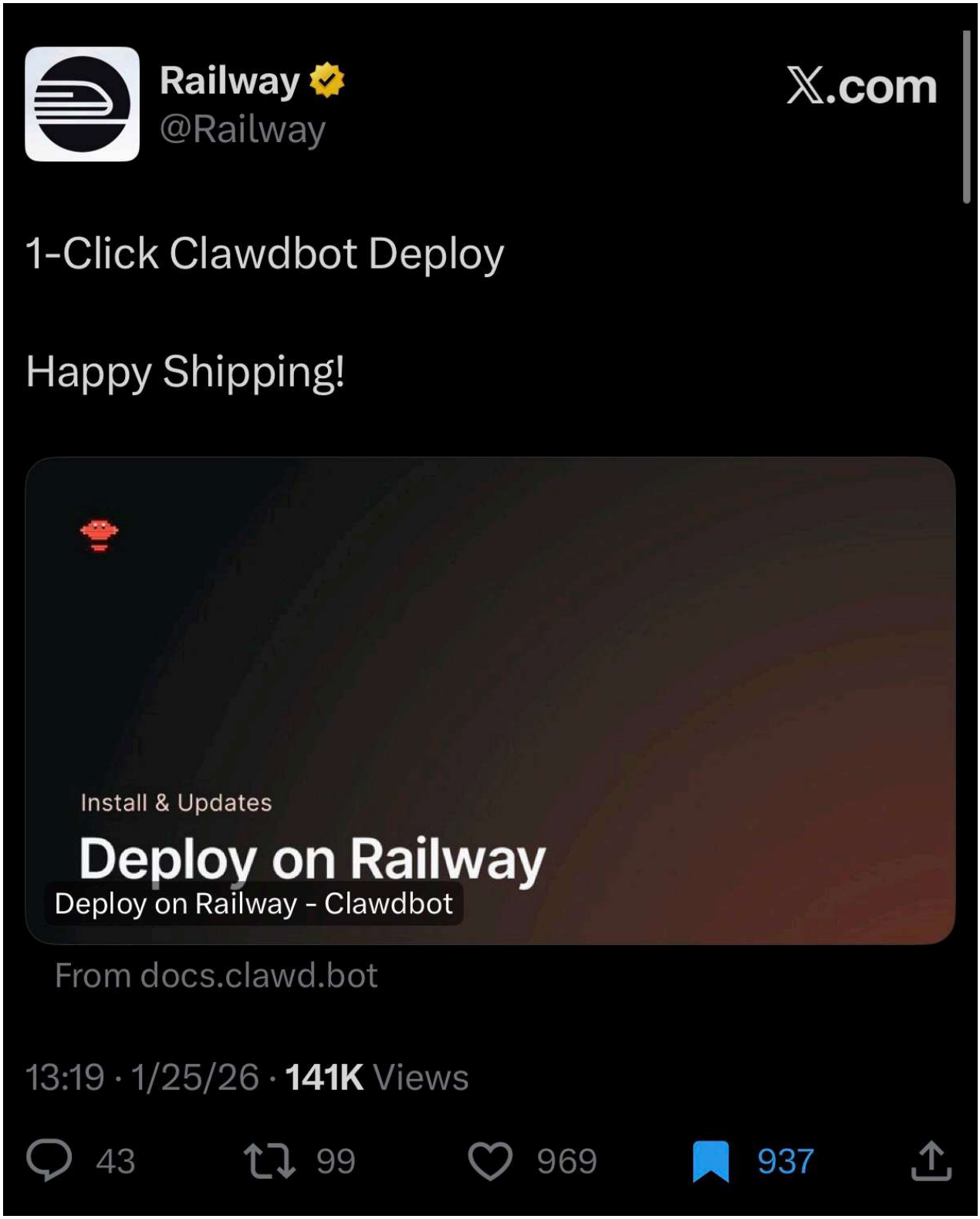
Local smoke test

```
docker build -t clawdbot-railway-template .

docker run --rm -p 8080:8080 \
  -e PORT=8080 \
  -e SETUP_PASSWORD=test \
  -e OPENCLAW_STATE_DIR=/data/.openclaw \
  -e OPENCLAW_WORKSPACE_DIR=/data/workspace \
  -v $(pwd)/.tmpdata:/data \
```

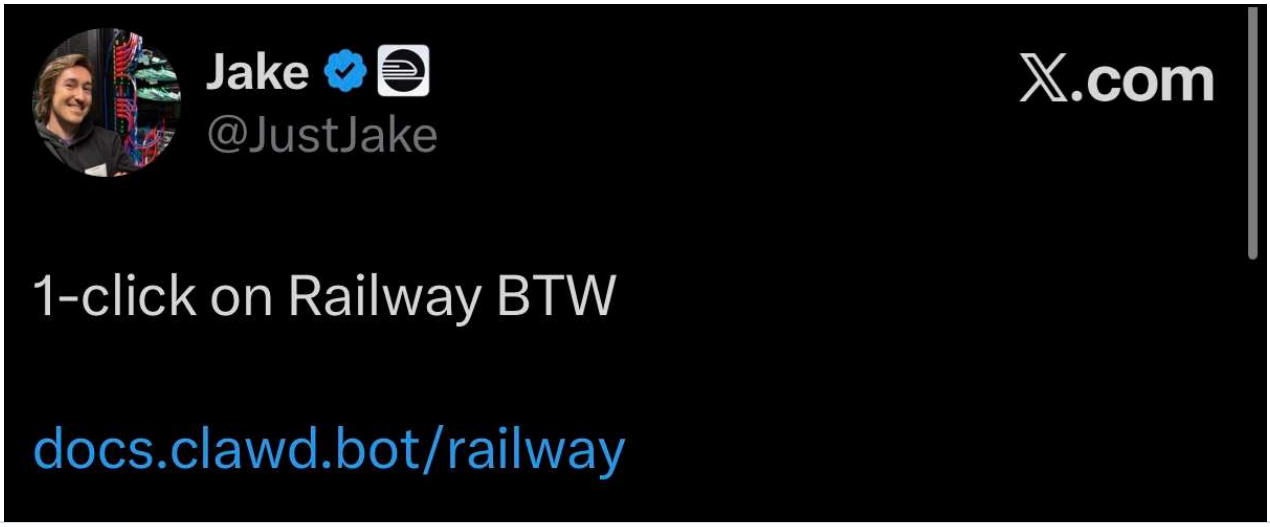
```
  clawdbot-railway-template

# open http://localhost:8080/setup (password: test)
```

Official template / endorsements

- Officially recommended by OpenClaw: https://docs.openclaw.ai/railway
- Railway announcement (official): Railway tweet announcing 1-click OpenClaw deploy



- Endorsement from Railway CEO: [Jake Cooper tweet endorsing the OpenClaw Railway template]



Releases

No releases published

Deployments 500+

- **renewed-kindness / production** last week
- **rpa-free-trial / production** last week
- **keen-connection / production** last month

+ more deployments

Packages

No packages published

Contributors 10

Languages

- JavaScript 96.4%
- Dockerfile 3.6%